

GREENHOUSE MONITORING USING ARDUINO

IoT Project
Report

Group introduction

Members:

1. Kausthubh Vishwanath
2. Medhya Parthakudi

Semester and Section: 2-F

Institute: RV Institute of Technology, JP Nagar

Report and presentation prepared by: Medhya Parthakudi

Prototype prepared by: Kausthubh Vishwanath

ABSTRACT

This project is based on monitoring temperature changes inside a greenhouse for proper growth of seedlings without any human interaction.

is important because any change in moisture and light intensities results in the death of crops and cause huge losses to farmers.

This system will keep the crops healthy by adjusting the environment as needed.

system will constantly measure the temperature and the soil's moisture level and if either of these change up to a certain level, it sends a message to the cultivator.

INTRODUCTION

The crops need enough light to grow well, and if there is less light, they may not grow properly.

The farmers have to control the water pump, light source, fan, and other devices at the same time to avoid these problems.

This system is created to simplify the tasks of farmers.

The system is fully automated and does not require any human intervention to operate. The users cell phone receives alerts when Arduino manages the process.

From 1800 to 2012, the number of people living in the world increased from 1 billion to 7 billion. It is anticipated to continue.

COMPONENTS

Hardware:

Arduino Uno-

Soil moisture sensors

DHT11 (Temperature and Humidity) sensor-

Light dependent resistor

COMPONENTS

Software:

Arduino Programming

Arduino Programming

WORKING

The system is a greenhouse system that has four sensors.

The data given to the micro controller is in the form of analog signals which is later transformed into digital form.

The information is displayed on the liquid crystal display.

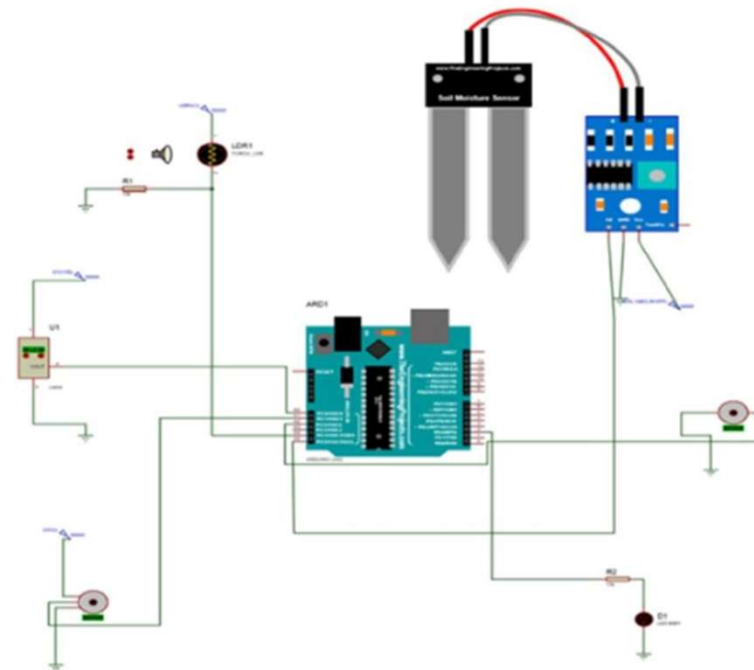
After the parameter values are observed, they can be adjusted.

The IOT greenhouse monitoring system utilizes a PC or phone base system.

This circuit uses a small computer to check and keep track of the temperature, humidity, soil moisture, and sunlight in the natural environment.

The system keeps track of the digital values of the different sensors at all times.

CIRCUIT DIAGRAM



CONCLUSION

Greenhouse monitoring is an essential aspect of modern agriculture, enabling farmers to optimize growing conditions and improve crop yields.

This project, which leverages the versatility and affordability of Arduino microcontrollers, exemplifies the power of DIY electronics in enhancing agricultural productivity and sustainability.

Achievements of the Project:

1. Soil Moisture Management:
2. Temperature and Humidity Control:
3. Light Intensity Regulation:
4. Automation and Efficiency:

REFERENCES

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